Merge script

2025-03-24

```
res_dir <- "../results/s01_qc_merge/"
```

This script merges our data with additional data from Madagascar, Borneo, Sumba, and 1000 Genomes Project, creating two datasets: one Malagasy only and another including Malagasy alongside all additional populations.

Generate gen tibble

```
if (file.exists("../data/raw/data_input/Madagascar_full.gt")) {
  madagascar <- gt_load("../data/raw/data_input/Madagascar_full.gt")
} else {
  madagascar <- gen_tibble("../data/raw/data_input/Madagascar_full.bed",
    quiet = TRUE,
    valid_alleles = c("T", "G", "A", "C", "I")
  )
}

cat("Number of individuals:", nrow(madagascar), "\n")
```

```
## Number of individuals: 192
```

```
cat("Number of loci:", count_loci(madagascar))
```

```
## Number of loci: 713014
```

```
if (file.exists("../data/raw/data_input/Madagascar_full_v2.bk")) {
  file.remove("../data/raw/data_input/Madagascar_full_v2.gt")
  file.remove("../data/raw/data_input/Madagascar_full_v2.bk")
  file.remove("../data/raw/data_input/Madagascar_full_v2.rds")
}
```

Our raw dataset has 713014 SNPs and 192 individuals

Add metadata

```
# Read in Pierron Malagasy metadata
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

# Add "RR" to pop_table$IID
pop_table$IID <- paste0(pop_table$IID, "RR")

# Find Malagasy entries
```

```
pop_table_sub <-
  subset(
    pop_table,
    pop_table$IID %in% madagascar$id & pop_table$V5 == "Madagascar"
  )

# match up the ID's with pop table metadata
matching_m <- match(madagascar$id, pop_table_sub$IID)

madagascar$population <-
  ifelse(!is.na(matching_m),
    pop_table_sub$Population[matching_m], madagascar$population
  )
```

Remove the "I" loci

```
remove <- which(show_loci(madagascar)$allele_ref == "I")
madagascar <- madagascar %>% select_loci(-all_of(c(remove)))
count_loci(madagascar)
```

```
## [1] 713013
```

Update build

```
new_bim_no_na <- read.table("../data/intermediate/Madagascar_full_HG38.bim",
  header = FALSE
)
colnames(new_bim_no_na) <- c(
  "chromosome", "snp_id", "genetic_dist",
  "physical_pos", "allele1", "allele2"
)

nrow(new_bim_no_na)
```

```
## [1] 692673
```

```
new_bim_no_na <- as_tibble(new_bim_no_na)

# subset to only autosomes
new_bim_no_na <- new_bim_no_na %>% filter(chromosome %in% c(1:22))

# order by chromosome
new_bim_no_na$chromosome <- as.integer(new_bim_no_na$chromosome)
new_bim_no_na <- new_bim_no_na %>% arrange(chromosome, physical_pos)
```

```
# select only the loci with updated positions
madagascar <-
  madagascar %>% select_loci_if(loci_names(genotypes) %in% new_bim_no_na$snp_id)

# find matching indices
match_indices <- match(show_loci(madagascar)$name, new_bim_no_na$snp_id)
matching_rows <- !is.na(match_indices)
```

```
# update position in gt
show_loci(madagascar)$position[matching_rows] <-
  new_bim_no_na$physical_pos[match_indices[matching_rows]]

# update chromosome in gt
show_loci(madagascar)$chromosome <-
  as.character(show_loci(madagascar)$chromosome)
show_loci(madagascar)$chromosome[matching_rows] <-
  new_bim_no_na$chromosome[match_indices[matching_rows]]

# update chr_int in gt
show_loci(madagascar)$chr_int <- as.integer(show_loci(madagascar)$chromosome)
count_loci(madagascar)
```

```
## [1] 675037
```

```
# update genetic distance in gt
show_loci(madagascar)$genetic_dist[matching_rows] <-
  new_bim_no_na$genetic_dist[match_indices[matching_rows]]

# remove entries with genetic distance == 0
no_dist <- which(show_loci(madagascar)$genetic_dist == 0)
madagascar <- madagascar %>% select_loci(-all_of(no_dist))

testthat::expect_false(is_loci_table_ordered(madagascar,
  error_on_false = FALSE,
  ignore_genetic_dist = TRUE
))
```

Malagasy Pierron data

```
if (file.exists("../data/raw/data_input/Pierron_Malagasy.gt")) {
  madagascar_p <- gt_load("../data/raw/data_input/Pierron_Malagasy.gt")
} else {
  madagascar_p <-
    gen_tibble("../data/raw/data_input/Pierron_Malagasy.bed", quiet = TRUE)
}

cat("Number of individuals:", nrow(madagascar_p), "\n")
```

```
## Number of individuals: 69
```

```
cat("Number of loci:", count_loci(madagascar_p))
```

```
## Number of loci: 661324
```

Append population info

```
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")
```

```
pop_table_sub <-
  subset(
    pop_table,
    pop_table$IID %in% madagascar_p$id & pop_table$V5 == "Madagascar"
  )

matching_m <- match(madagascar_p$id, pop_table_sub$IID)
madagascar_p$population <- ifelse(!is.na(matching_m),
  pop_table_sub$Population[matching_m],
  madagascar_p$population
)
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_merge.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge.rds")
}
```

```
## [1] TRUE
```

Merge the two datasets

```
malagasy_merged <- rbind(madagascar, madagascar_p,
  as_is = FALSE, flip_strand = TRUE,
  quiet = FALSE, backingfile = "../data/intermediate/rbind_data/malagasy_merge"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 672886 reduced to 628374
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 661324 reduced to 628374
## ( 125 were flipped to match the reference set)
## ( 0 are ambiguous, of which 0 were removed)
##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mala
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## make sure that you do NOT delete those files!
## to reload the gen_tibble in another session, use:
## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge_v2.bk")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge_v2.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_v2.rds")
}
```

```
## [1] TRUE
```

Sort the order

```
testthat::expect_false(is_loci_table_ordered(malagasy_merged,
  error_on_false = FALSE
))
# reorder the loci
malagasy_merged <- gt_order_loci(malagasy_merged,
  use_current_table = FALSE,
  ignore_genetic_dist = TRUE
)
```

```
## Genetic distances are not sorted, setting them to zero
```

```
##
## gen_backing files updated, now
```

```
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
```

```
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
```

```
## make sure that you do NOT delete those files!
```

Quality control on the merged dataset

First we create a gt 'malagasy_merge_clean_no_maf_1st_degree' for ROH analysis, this data will include all but first degree relatives and is not MAF filtered.

```
malagasy_merged <- malagasy_merged %>% group_by(population)
loci_report_m <- qc_report_loci(malagasy_merged)
malagasy_merged <- malagasy_merged %>% ungroup()
```

Filter loci

```
to_keep <- which(loci_report_m$hwe_p > 0.01)
malagasy_merged <- malagasy_merged %>% select_loci(all_of(to_keep))
count_loci(malagasy_merged)
```

```
## [1] 624099
```

```
malagasy_merged <-
  malagasy_merged %>% select_loci_if(loci_missingness(genotypes) < 0.01)
count_loci(malagasy_merged)
```

```
## [1] 606305
```

```
malagasy_merged <-
  malagasy_merged %>% select_loci_if(show_loci(genotypes)$chromosome %in% c(1:22))
count_loci(malagasy_merged)
```

```
## [1] 606305
```

Filter individuals

```
malagasy_merged <- malagasy_merged %>% group_by(population)
individual_report_m <- qc_report_indiv(malagasy_merged, kings_threshold = 0.177)
malagasy_merged <- malagasy_merged %>% ungroup()
malagasy_merged <-
  malagasy_merged %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TRUE)
malagasy_merged <-
  malagasy_merged %>% filter(indiv_missingness(genotypes) < 0.1)
```

Loci and sample count

```
nrow(malagasy_merged)
```

```
## [1] 243
```

```
count_loci(malagasy_merged)
```

```
## [1] 606305
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.rds")
}
```

```
## [1] TRUE
```

```
gt_save(malagasy_merged,
  file_name = "../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.gt
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## make sure that you do NOT delete those files!
## to reload the gen_tibble in another session, use:
## gt_load('../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.gt')
## [1] "../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge_v2.
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge_v2.
```

```
gt_update_backingfile(malagasy_merged,
  backingfile = "../data/intermediate/rbind_data/malagasy_merge_clean_no_maf_1st_degree"
)
```

```
## Genetic distances are not sorted, setting them to zero
##
## gen_backing files updated, now
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
```

```
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag

## make sure that you do NOT delete those files!

## # A gen_tibble: 606305 loci
## # A tibble:     243 x 3
##    id       population     genotypes
##    <chr>    <chr>         <vctr_SNP>
##  1 BET001RR Betsimisaraka          1
##  2 BET002RR Betsimisaraka          2
##  3 BET003RR Betsimisaraka          3
##  4 BET004RR Betsimisaraka          4
##  5 BET005RR Betsimisaraka          5
##  6 BET006RR Betsimisaraka          6
##  7 BET007RR Betsimisaraka          7
##  8 BET009RR Betsimisaraka          8
##  9 BET010RR Betsimisaraka          9
## 10 BET011RR Betsimisaraka         10
## # i 233 more rows
```

Next we add MAF filtering and remove second degree relatives to create gt 'malagasy_merge_clean' for other analysis.

Add MAF filtering

```
malagasy_merged <-
  malagasy_merged %>% select_loci_if(loci_maf(genotypes) > 0.025)
count_loci(malagasy_merged)
```

```
## [1] 579742
```

Add second degree relatedness filtering

```
malagasy_merged <- malagasy_merged %>% group_by(population)
individual_report_m <-
  qc_report_indiv(malagasy_merged, kings_threshold = 0.0884)
malagasy_merged <- malagasy_merged %>% ungroup()
malagasy_merged <-
  malagasy_merged %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TRUE)
nrow(malagasy_merged)
```

```
## [1] 233
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge_clean.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean.rds")
}
```

```
## [1] TRUE
```

```
gt_save(malagasy_merged,
  file_name = "../data/intermediate/rbind_data/malagasy_merge_clean"
)
```

```
##
```

```
## gen_tibble saved to ../data/intermediate/rbind_data/malagasy_merge_clean.gt
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## make sure that you do NOT delete those files!
## to reload the gen_tibble in another session, use:
## gt_load('../data/intermediate/rbind_data/malagasy_merge_clean.gt')
## [1] "../data/intermediate/rbind_data/malagasy_merge_clean.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge_v2.
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge_v2.
```

```
malagasy_merged <-
  gt_update_backingfile(malagasy_merged,
    backingfile = "../data/intermediate/rbind_data/malagasy_merge_clean"
  )
```

```
## Genetic distances are not sorted, setting them to zero
##
## gen_backing files updated, now
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## make sure that you do NOT delete those files!
```

Impute

```
malagasy_merged <- gt_impute_simple(malagasy_merged, method = "mean")
```

LD clumping

```
to_keep_LD <- loci_ld_clump(malagasy_merged,
  thr_r2 = 0.2, size = 100,
  return_id = TRUE, use_positions = TRUE
)
malagasy_merged_ld <- malagasy_merged %>% select_loci(all_of(to_keep_LD))
```

Loci and sample count

```
nrow(malagasy_merged_ld)
```

```
## [1] 233
```

```
count_loci(malagasy_merged_ld)
```

```
## [1] 223963
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_merge_clean_ld.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_ld.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_ld.bk")
```

```
  file.remove("../data/intermediate/rbind_data/malagasy_merge_clean_ld.rds")
}
```

```
## [1] TRUE
```

Save LD filtered dataset

```
gt_save(malagasy_merged_ld,
  file_name = "../data/intermediate/rbind_data/malagasy_merge_clean_ld"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/malagasy_merge_clean_ld.gt
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## make sure that you do NOT delete those files!
## to reload the gen_tibble in another session, use:
## gt_load('../data/intermediate/rbind_data/malagasy_merge_clean_ld.gt')
## [1] "../data/intermediate/rbind_data/malagasy_merge_clean_ld.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge_clea
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_merge_clea
```

```
malagasy_merged <-
  gt_update_backingfile(malagasy_merged_ld,
    backingfile = "../data/intermediate/rbind_data/malagasy_merge_clean_ld"
  )
```

```
## Genetic distances are not sorted, setting them to zero
##
## gen_backing files updated, now
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## make sure that you do NOT delete those files!
```

Load Borneo data

```
if (file.exists("../data/raw/data_input/Borneo_raw_data.gt")) {
  borneo <- gt_load("../data/raw/data_input/Borneo_raw_data.gt")
} else {
  borneo <-
    gen_tibble("../data/raw/data_input/Borneo_raw_data.bed",
      valid_alleles = c("A", "T", "C", "G", "I", "N"),
      quiet = TRUE
    )
}

borneo$population <- "Borneo"
```

```
if (file.exists("../data/intermediate/rbind_data/malagasy_borneo.gt")) {
  file.remove("../data/intermediate/rbind_data/malagasy_borneo.gt")
  file.remove("../data/intermediate/rbind_data/malagasy_borneo.bk")
  file.remove("../data/intermediate/rbind_data/malagasy_borneo.rds")
}
```

```
## [1] TRUE
```

Merge

```
malagasy_merged <-
  gt_load("../data/intermediate/rbind_data/malagasy_merge_clean.gt")
malagasy_borneo <- rbind(malagasy_merged, borneo,
  as_is = FALSE, flip_strand = TRUE,
  quiet = FALSE, backingfile = "../data/intermediate/rbind_data/malagasy_borneo"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ----------------------------
## dataset: reference
## number of SNPs: 579742 reduced to 510970
## ( 0 are ambiguous, of which 0  were removed)
## ----------------------------
## dataset: target
## number of SNPs: 716503 reduced to 510970
## ( 93610 were flipped to match the reference set)
## ( 1816 are ambiguous, of which 1816 were removed)
##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mala
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malag
## make sure that you do NOT delete those files!
## to reload the gen_tibble in another session, use:
## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/malagasy_borne
```

Load 1000 Genomes data

Cutting down to populations of interest

```
if (file.exists("../data/raw/data_input/1000_Genomes_b37_clean.gt")) {
  g1000 <- gt_load("../data/raw/data_input/1000_Genomes_b37_clean.gt")
} else {
  g1000 <-
    gen_tibble("../data/raw/data_input/1000_Genomes_b37_clean.bed",
      quiet = TRUE
    )
}

# append population info
```

```
pop_table <- read.table("../data/raw/data_input/Madagascar_Borneo.poptable",
  sep = "\t", fill = TRUE
)
colnames(pop_table) <- c("IID", "V2", "FID", "Population", "V5", "V6", "V7")

pop_table_sub <- subset(pop_table, pop_table$IID %in% g1000$id)

matching_m <- match(g1000$id, pop_table_sub$IID)
g1000$population <- ifelse(!is.na(matching_m),
  pop_table_sub$V2[matching_m],
  g1000$population
)

pop <- c("KHV", "BEB", "YRI", "PJL", "GWD", "MSL", "ESN", "LWK", "CHS")
#- Kinh in Ho Chi Minh City, Vietnam
#- Bengali in Bangladesh
#- Han Chinese South
#- Luhya in Webuye, Kenya
#- Yoruba in Ibadan, Nigeria
#- Punjabi in Lahore, Pakistan
#- Gambian in Western Division, The Gambia - Mandinka
#- Mende in Sierra Leone
#- Esan in Nigeria

g1000 <- g1000 %>% filter(population %in% pop)

count_loci(g1000)
```

```
## [1] 670552
```

```
nrow(g1000)
```

```
## [1] 882
```

Merge

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000.rds")
}
```

```
## [1] TRUE
```

```
mad_bor_g1000 <- rbind(malagasy_borneo, g1000,
  as_is = FALSE, flip_strand = TRUE,
  quiet = FALSE, backingfile = "../data/intermediate/rbind_data/mad_bor_g1000"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 510970 reduced to 502627
## ( 0 are ambiguous, of which 0  were removed)
```

```
## ----------------------------
## dataset: target
## number of SNPs: 670552 reduced to 502627
## ( 91215 were flipped to match the reference set)
## ( 3 are ambiguous, of which 3 were removed)
##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
## make sure that you do NOT delete those files!
## to reload the gen_tibble in another session, use:
## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000.g
```

Load Sumba data

```
if (file.exists("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.gt")) {
  sumba <- gt_load("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.gt")
} else {
  sumba <-
    gen_tibble("../data/raw/data_input/GSE76645_INDONESIA_SUMBA_TIMOR.bed",
      quiet = TRUE
    )
}

sumba$population <- "Sumba"

timor <- read.table("../data/raw/data_input/timor")
timor$population <- "Timor"

matching_m <- match(sumba$id, timor$V2)
sumba$population <- ifelse(!is.na(matching_m),
  timor$population[matching_m],
  sumba$population
)

count_loci(sumba)
```

```
## [1] 713246
```

```
nrow(sumba)
```

```
## [1] 240
```

```
set.seed(1)

subsample <- function(x) {
  if (nrow(x) <= 30) {
    return(x)
  } else {
    sample_n(x, size = 30)
  }
```

```
}

sumba <- sumba %>%
  group_by(population) %>%
  group_modify(~ subsample(.x)) %>%
  ungroup()

nrow(sumba)
```

```
## [1] 37
```

Merge

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum.rds")
}
```

```
## [1] TRUE
```

```
mad_bor_g1000_sum <- rbind(mad_bor_g1000, sumba,
  as_is = FALSE,
  flip_strand = TRUE,
  quiet = FALSE,
  backingfile = "../data/intermediate/rbind_data/mad_bor_g1000_sum"
)
```

```
## harmonising loci between two datasets
## flip_strand =  TRUE  ; remove_ambiguous =  TRUE
## ------------------------------
## dataset: reference
## number of SNPs: 502627 reduced to 502609
## ( 0 are ambiguous, of which 0  were removed)
## ------------------------------
## dataset: target
## number of SNPs: 713246 reduced to 502609
## ( 91213 were flipped to match the reference set)
## ( 2056 are ambiguous, of which 2056 were removed)
##
## gen_tibble saved to /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
## make sure that you do NOT delete those files!
## to reload the gen_tibble in another session, use:
## gt_load('/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_
```

Quality control on the merged dataset

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
loci_report_m <- qc_report_loci(mad_bor_g1000_sum)
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% ungroup()
```

Filter loci

```
to_keep <- which(loci_report_m$hwe_p > 0.01)
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% select_loci(all_of(to_keep))
count_loci(mad_bor_g1000_sum)
```

```
## [1] 499993
```

```
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% select_loci_if(loci_missingness(genotypes) < 0.01)
count_loci(mad_bor_g1000_sum)
```

```
## [1] 499354
```

```
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% select_loci_if(loci_maf(genotypes) > 0.025)
count_loci(mad_bor_g1000_sum)
```

```
## [1] 498926
```

```
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% select_loci_if(show_loci(genotypes)$chromosome %in% c(1:22))
count_loci(mad_bor_g1000_sum)
```

```
## [1] 498926
```

Filter indiv

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
individual_report_m <-
  qc_report_indiv(mad_bor_g1000_sum, kings_threshold = 0.0884)
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% ungroup()
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% filter(id %in% individual_report_m$id & individual_report_m$to_keep == TRUE)
mad_bor_g1000_sum <-
  mad_bor_g1000_sum %>% filter(indiv_missingness(genotypes) < 0.1)
nrow(mad_bor_g1000_sum)
```

```
## [1] 1178
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 498926
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 1178
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 498926
```

Save

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.rds")
}
```

```
## [1] TRUE
```

```
gt_save(
  mad_bor_g1000_sum,
  "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt
```

```
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
```

```
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
```

```
## make sure that you do NOT delete those files!
```

```
## to reload the gen_tibble in another session, use:
```

```
## gt_load('../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt')
```

```
## [1] "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum.
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum.
```

```
mad_bor_g1000_sum <-
  gt_update_backingfile(mad_bor_g1000_sum,
    backingfile = "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean"
  )
```

```
## Genetic distances are not sorted, setting them to zero
```

```
##
## gen_backing files updated, now
```

```
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
```

```
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
```

```
## make sure that you do NOT delete those files!
```

Impute

```
mad_bor_g1000_sum <- gt_impute_simple(mad_bor_g1000_sum, method = "mean")
```

LD clumping

```
if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld.gt")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld.bk")
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld.rds")
}
```

```
## [1] TRUE
```

```
to_keep_LD <- loci_ld_clump(mad_bor_g1000_sum,
  thr_r2 = 0.2, size = 100,
  return_id = TRUE, use_positions = TRUE
)

mad_bor_g1000_sum_ld <- mad_bor_g1000_sum %>% select_loci(all_of(to_keep_LD))
```

Loci and sample count

```
nrow(mad_bor_g1000_sum_ld)
```

```
## [1] 1178
```

```
count_loci(mad_bor_g1000_sum_ld)
```

```
## [1] 188396
```

Save LD filtered dataset

```
gt_save(
  mad_bor_g1000_sum_ld,
  "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld.gt

## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_l

## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_l

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld.gt')

## [1] "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_
```

```
mad_bor_g1000_sum_ld <-
  gt_update_backingfile(mad_bor_g1000_sum_ld,
    backingfile = "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_ld"
  )
```

```
## Genetic distances are not sorted, setting them to zero
```

```
##
## gen_backing files updated, now
## using bigSNP file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
## with backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_
## make sure that you do NOT delete those files!
```

Methods

Genetic and phenotypic data collected by the team of Dr George Perry at Pennsylvania State University with the University of Antananarivo. Dr Jason Hodgson and Dr Rindra Rakotoarivony led this data collection in 2015. These data comprise of 192 individuals from five regions: Merina people from the Central Highlands, Betsileo people from the southern Highlands, Sakalava and Tsimihety people from the northwestern coast, Betsimisaraka people from the eastern coast, and people of various ethnic groups from the extreme north in the town of Diego. Samples were collected from married couples, and ~690,000 SNPs were typed using the Illumina HumanOmniExpress-24 array.

Two main datasets were created, a Malagasy only dataset and a global dataset combining Malagasy samples with comparative data. The Malagasy dataset was created by merging the data from this study with Malagasy data from Pierron et al. (2014) (n = 69). For a comparative global dataset, this data was combined with data from southeast Asian and African populations in the 1000 Genomes Project Consortium (2015) dataset (n = 882), individuals from Borneo from Brucato et al. (2016) (n = 39), and individuals from the islands of Sumba (n = 30, subsampled at random from 37) and Timor (n = 7) from Cox et al. (2016).

Scripts in R were used to map data to GRCH38 from build GRCH37. The package BiomaRt (version '2.56.1'), which utilises the ensembl database, was employed to do so (Durinck et al., 2009, 2005). Data were then cleaned and merged in R with package tidypopgen (Carter and Manica, 2025). Only autosomal SNPs were analysed. All SNPs with a missingness rate above 0.01 across all samples were removed. All individuals with a genotype call rate less than 0.9 (missingness exceeding 0.1) were removed. All SNPs with minor allele frequency below a threshold of 0.025 were removed. All SNPs with a Hardy-Weinberg equilibrium test p-value below 0.01 were removed. Close relatives were removed using the KING robust kinship coefficient (Manichaikul et al., 2010), implemented in tidypopgen. One individual was removed from each pair with a kinship coefficient greater than 0.0884, equivalent to removing relatives up to the second degree. After this filtering, the Malagasy dataset consisted of 233 individuals and 579742 SNPs, and the global dataset consisted of 1178 individuals and 498926 SNPs.

SNPs in linkage were then removed by LD clumping (Privé et al., 2018), using tidypopgen. Those with an r2 value greater than 0.2 in a sliding window of 100 kilobases were removed. After pruning, the Malagasy dataset consisted of 223963 SNPs and the global dataset consisted of 188396 SNPs.